

Iker Garcia Morales

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Electrical Engineering and Robotics student with experience in processor design, RTL development, CMOS VLSI layout, and hardware verification. Interested in CPU architecture, ASIC design, and open-source hardware.

SKILLS:

Hardware Description Languages: SystemVerilog, Verilog, VHDL

Programming: C, C++, Python, MATLAB

Processor & Digital Design: ISA Design, RTL Design, Multicycle Datapaths, FSMs, Digital Logic

VLSI & ASIC Design: CMOS Schematic and Layout, DRC, ERC, NCC, ASIC Design Flow

Simulation & Tools: Electric VLSI, ModelSim, Quartus Prime, Yosys, Git, LTspice

Embedded Systems: Arduino, ESP32, STM32.

Other: Linux, Simulink

EDUCATION:

John Brown University, Siloam Springs, AR

Bachelor of Science in Electrical Engineering, Double Major in Robotics and Mechatronics Engineering.

Expected May 2027

Relevant Coursework

Digital IC Design, Digital Electronics, Embedded Systems, Electronics, Electrical Circuits, Control Systems.

PROJECTS:

Silicio-16: Custom 16-bit Processor (SystemVerilog)

Designing a custom 16-bit multicycle processor with eight general-purpose registers and arithmetic, logical, memory, branch, comparison, and jump instructions. Defined the ISA, instruction formats, datapath, and execution cycles.

Currently implementing the ALU, register file, program counter, instruction register, immediate-extension logic, and FSM-based control unit.

ASIC Design: 8-bit MIPS Processor (Electric VLSI)

Designed an 8-bit MIPS-style processor using a bit-slice architecture, including the datapath, ALU, register structures, and FSM-based control unit. Completed full-custom CMOS schematic and layout, passed DRC, ERC, and NCC verification, simulated the processor using IRSIM, and integrated the core into a MOSIS TinyChip padframe.

ASIC Design: Similarity Detector (Electric VLSI)

Designed an 8-bit similarity detector using an XNOR array, popcount circuit, and programmable threshold comparator.

Completed functional simulation, transistor-level schematic, CMOS layout, and DRC, ERC, and NCC verification.

Digital Electronics: Serial 4-bit ALU

Designed and simulated a gate-level serial ALU in Quartus Prime supporting addition, subtraction, XOR, and increment operations through shift registers and sequential carry logic.

RELEVANT WORK EXPERIENCE:

John Brown University — Siloam Springs, AR

Teaching Assistant – Digital Electronics

Jan 2025 – Present

Facilitate digital circuit design and troubleshooting labs, support students working with logic gates, flip-flops, multiplexers, and counters, and evaluate assignments for circuit correctness and technical precision.

Learning Coach and Engineering Tutor – Student Support Services

Sep 2024 – Present

Provide one-on-one and small-group support in calculus, electrical circuits, technical problem-solving, and study strategies for engineering students.

LEADERSHIP EXPERIENCE:

IEEE Student Branch – Junior Senator

Student Government Association – Freshman Senator, Junior Senator.

Walton International Scholarship Scholar.